Distributing Objects into Boxes

Many counting problems can be solved by enumerating the different ways objects can be placed into boxes where the order these objects are placed into the boxes does not matter.

The objects and the boxes can be either *Distinguishable* (labeled) or *Indistinguishable* (unlabeled).

Examples:

- D-D: How many ways to place n labeled objects in m labeled boxes
- D-I: How many ways to place n labeled objects in m unlabeled boxes
- I-D: How many ways to place n unlabeled objects in m labeled boxes
- I-I: How many ways to place n unlabeled objects in m unlabeled boxes